

Hassan Alshabanh

Portfolio Website

Assignment 4 – Final Presentation

STUDENT

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KFUPM

COURSE

ICS Department

STACK

HTML · CSS · JavaScript

Project Overview & Objectives

What is this project?

A fully responsive personal portfolio website built from scratch using pure HTML5, CSS3, and Vanilla JavaScript — no frameworks, no libraries.

Personal Motivation

I wanted a real place to show my work — not a template, but something I built myself and understand completely. Each assignment added a new layer of skill on top of the last.

OBJECTIVES

- › Demonstrate HTML, CSS & JavaScript mastery
- › Integrate two live external APIs
- › Build responsive, accessible design
- › Apply professional code quality & structure
- › Deploy a public, shareable web application

Technical Demonstration



Dark Mode

localStorage persistence across sessions



Live Weather API

Open-Meteo — temperature, wind, humidity for KFUPM



Quote API

Advice Slip with 5 hardcoded fallbacks on failure



Search + Filter + Sort

All three controls work simultaneously on projects



Skill Bar Animation

IntersectionObserver triggers once on scroll



Form Validation

Empty-field + regex email check with styled feedback

Technical Architecture & AI Integration

Architecture

<code>index.html</code>	Semantic HTML5 structure
<code>css/styles.css</code>	Flexbox + dark mode + animations
<code>js/script.js</code>	All interactivity, APIs, observers
<code>assets/images/</code>	Lazy-loaded project images
<code>docs/</code>	AI report + technical documentation

AI Tools Used

Claude (Anthropic)

Debugging, code review, bug fixes, docs

ChatGPT

Explaining JS concepts & API structures

GitHub Copilot

Completing repetitive HTML structures

Key bug caught: `<section>` never fires submit events — form validation was silently broken until the listener was moved to `<form id='contactFormEl'>`.

Technical Deep Dive

✓ Resolved

- › Form validation silently broken — `<section>` doesn't fire submit; fixed by targeting `<form id='contactFormEl'>`
- › `id='Learning Goals'` had a space — invalid HTML; renamed to `id='learning-goals'`
- › Search bar appearing beside project cards — moved controls outside flex container, added `width: auto`
- › `showFormMessage()` depended on outer variable — refactored to accept `formEl` as parameter

🌟 Future Work

- › Add a backend to actually send contact form emails
- › Add more projects as skills grow
- › Implement smooth scroll animations between sections
- › Add multilingual support (Arabic / English toggle)

Conclusion

Outcomes & Achievements

- ✓ Fully responsive site deployed and publicly accessible
- ✓ Two live APIs with graceful error handling
- ✓ 9 JavaScript features, all tested across devices
- ✓ Professional code structure with clear documentation
- ✓ Used AI as a learning tool — not a shortcut

9

JS Features

2

Live APIs

100%

Vanilla JS

A3 → A4

Iterative Build

Thank you